

Package Insert

NORFOX TABLETS

COMPOSITION:

Norfloxacin 400mg per tablet.

INDICATION:

Indicated for the oral treatment of adults with complicated and uncomplicated urinary tract infections that are caused by susceptible strains of microorganisms. Consideration should be given to official guidance on the appropriate use of antibacterial agents.

PHARMACOLOGY:

Norfloxacin is a fluorinated 4 quinolone antimicrobial agent structurally related to nalidixic acid but with a wider antibacterial spectrum and greater activity. It has been shown to be bactericidal with the MIC being close to the MBC and to act by inhibiting the A subunit of DNA gyrase (topoisomerase) which is essential in the reproduction of bacterial DNA. The reported minimum inhibitory concentrations for susceptible bacteria range from 0.006 to 1.6 microgram per mL. Sensitive Gram negative organism includes *Escherichia coli*, *Citrobacter*, *Enterobacter*, *Klebsiella*, *Proteus*, *Salmonella*, *Shigella*, *Yersinia*, *Pseudomonas aeruginosa*, *Haemophilus*, *Neisseria*, *Acinetobacter*, *Campylobacter*, *Providencia*, *Serratia* and *Legionella*. Susceptible Gram positive organism includes *Staphylococci*, *Streptococci* and *Listeria*. Norfloxacin has little antifungal activity.

Maximum plasma concentration of about 1.5 microgram of Norfloxacin per ml have been achieved 1 to 2 hours after the administration of a 400 mg dose by mouth. Norfloxacin is 14% bound to plasma protein. The plasma half life is about 4 hours. 30 % is excreted unchanged in the urine within 24 hours and thus producing a high urinary concentration. Some metabolism occurs, possibly in the liver have been identified, some of which has antibacterial activity. Norfloxacin has been detected in bile, prostatic fluid, umbilical cord blood and amniotic fluid.

DOSAGE AND ADMINISTRATION:

Norfloxacin is administered orally, 1 hour before or 2 hours after a meal.

Urinary Tract Infections: Complicated or Uncomplicated: 400 mg twice daily

Gastrointestinal Infections: 400 mg twice daily

Gonorrhea: 800 mg single dose

Acute or chronic prostatitis: 400 mg twice daily

CONTRAINDICATION:

Norfloxacin is contraindicated in prepubertal children and growing adolescents. Norfloxacin should not be taken with Probenecid as it decreases the excretion of Norfloxacin in the urine. It is not recommended to take Norfloxacin with alkalinising agents that causes urine to be alkaline and thus reducing the excretion of Norfloxacin. Norfloxacin is contraindicated in individuals with known hypersensitivity to Norfloxacin and/or other related quinolone antimicrobial agents.

PREGNANCY AND LACTATION:

There are no adequate and controlled studies to date using norfloxacin in pregnant women. Since the drug, like most other quinolones, causes arthropathy in immature animals, norfloxacin should not be used in pregnant or nursing women.

WARNINGS AND PRECAUTIONS:

- a. The use of Norfloxacin should be avoided in patients who have experienced serious adverse reactions in the past when using fluoroquinolones containing products (see section Adverse Effects/Undesirable Effects). Treatment of these patients with [INN] should only be initiated in the absence of alternative treatment options and after careful benefit/risk assessment.
- b. Norfloxacin should be given with caution to patients with damaged central nervous system, to patients subjected to convulsions and patients with impaired renal and / or hepatic functions.
- c. In patients with impaired renal function, Norfloxacin should not be used for more than 8 weeks and in patients with creatinine clearance of less than 10ml/min, it is recommended that the usual dose to be halved and the dosage time interval be doubled.
- d. Phototoxicity may occur. To take precautionary measures against direct exposure to sunlight.
- e. It is recommended to maintain adequate fluid intake during treatment with Norfloxacin to avoid risk of crystalluria.
- f. Exacerbation of myasthenia gravis. Fluoroquinolones have neuromuscular blocking activity and may exacerbate muscle weakness in person with myasthenia gravis. Post marketing serious adverse events,

- including deaths and requirement for ventilator support have been associated with fluoroquinolones use in persons with myasthenia gravis. Avoid fluoroquinolones in patients with known history of myasthenia gravis.
- g. Aortic aneurysm and dissection. Epidemiologic studies report an increased risk of aortic aneurysm and dissection after intake of fluoroquinolones, particularly in the older population. Therefore, fluoroquinolones should only be used after careful benefit-risk assessment and after consideration of other therapeutic options in patients with positive family history of aneurysm disease, or in patients diagnosed with pre-existing aortic aneurysm and/or aortic dissection, or in presence of other risk factors or conditions predisposing for aortic aneurysm and dissection (e.g. Marfan syndrome, vascular Ehlers-Danlos syndrome, Takayasu arteritis, giant cell arteritis, Behcet's disease, hypertension, known atherosclerosis). In case of sudden abdominal, chest or back pain, patients should be advised to immediately consult a physician in an emergency department.
- h. Prolonged, disabling and potentially irreversible serious adverse drug reactions. Very rare cases of prolonged (continuing months or years), disabling and potentially irreversible serious adverse drug reactions affecting different, sometimes multiple body systems (musculoskeletal, nervous, psychiatric and senses) have been reported in patients receiving fluoroquinolones irrespective of their age and pre-existing risk factors. Norfloxacin should be discontinued immediately at the first signs or symptoms of any serious adverse reaction and patients should be advised to contact their prescriber for advice.
- i. Tendinitis and tendon rupture. Tendinitis and tendon rupture (especially but not limited to Achilles tendon), sometimes bilateral, may occur as early as within 48 hours of starting treatment with fluoroquinolones and have been reported to occur even up to several months after discontinuation of treatment. The risk of tendinitis and tendon rupture is increased in older patients (above 60 years of age), with renal impairment, patients with solid organ transplants, and those treated concurrently with corticosteroids*. Therefore, concomitant use of corticosteroids should be avoided. At the first sign of tendinitis (e.g. painful swelling, inflammation) the treatment with Norfloxacin should be discontinued and alternative treatment should be considered. The affected limb(s) should be appropriately treated (e.g. immobilisation). Corticosteroids should not be used if signs of tendinopathy occur.
- j. Peripheral neuropathy. Cases of sensory or sensorimotor polyneuropathy resulting in paraesthesia, hypaesthesia, dysesthesia, or weakness have been reported in patients receiving quinolones and fluoroquinolones. Patients under treatment with [INN] should be advised to inform their doctor and pharmacist prior to continuing treatment if symptoms of neuropathy such as pain, burning, tingling, numbness, or weakness develop in order to prevent the development of potentially irreversible condition (see section Adverse Effects/Undesirable Effects).

SIDE EFFECTS:

Common side effects include nausea, vomiting, diarrhoea, rash, heartburn, dizziness, headache and abdominal pain/ cramps. On rare occasions, abnormalities have been observed in laboratory values, including leucopenia, eosinophilia and increased SGOT, SGPT, alkaline phosphate, bilirubin BUN and creatinine values. Post-marketing experience: exacerbation of myasthenia gravis.

Musculoskeletal and connective tissue disorders*, Nervous system disorders*, General disorders and administrative site conditions*, Psychiatric disorders*, Eye disorders*, Ear and labyrinth disorders*.

*Very rare cases of prolonged (up to months or years), disabling and potentially irreversible serious drug reactions affecting several, sometimes multiple, system organ classes and senses (including reactions such as tendinitis, tendon rupture, arthralgia, pain in extremities, gait disturbance, neuropathies associated with paraesthesia, depression, fatigue, memory impairment, sleep disorders, and impairment of hearing, vision, taste and smell) have been reported in association with the use of fluoroquinolones in some cases irrespective of pre-existing risk factors (see section Warnings and Precautions).

DRUG INTERACTIONS:

Antacids containing magnesium hydroxide or aluminum hydroxide may decrease absorption of oral norfloxacin, and the drugs probably should not be administered concomitantly. Initiation of oral norfloxacin therapy in patients stabilized on warfarin has resulted in prolongation of the prothrombin time in several patients. Oral multivitamin and mineral supplements containing divalent or trivalent cations such as iron or zinc may interfere with oral absorption of norfloxacin resulting in decreased serum and urine concentrations of the quinolone. Therefore, these multivitamins and/or mineral supplements should not be ingested concomitantly with or within 2 hours of a norfloxacin dose. Concomitant use of some quinolones in patients receiving theophylline has resulted in increased plasma theophylline concentrations and decreased clearance of the drug and may increase the risk of theophylline-related adverse effects.

SYMPTOMS AND TREATMENT OF OVERDOSAGE:

No experience is available in overdosage, consequently, the signs and symptoms and treatment are unknown. In the event of an acute overdosage, the stomach should be emptied by inducing vomiting or by stomach pumping, the patient should be kept under close observation and symptomatic and supportive treatment should be administered. The patient should be sufficiently hydrated as the intake of high doses of Norfloxacin may cause crystalluria.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINERIES:

Norfloxacin may alter a patient's reactivity so that the ability to drive, operate machinery or work without firm support is impaired, especially at the start of treatment, on increasing the dosage or when switching medication and in conjunction with alcohol.

STORAGE:

Store below 30° C and protect from light. Keep out of reach of children.

SHELF-LIFE:

3 years.

PRESENTATION:

Available in strips of 10 tablets. Packing in box of 100's.

MANUFACTURER:

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PRODUCT REGISTRATION HOLDER:

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